Sensitivity analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask what changed, what stayed fixed and whether correctness was actually measured.

Teach sensitivity with honest aggregation, label and selection limits.

Keep one habit: Model retrieval summaries → trace cutoff equality → compare hypothetical labels → preserve all settings → assess independent range.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **sensitivity analysis**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of ablation studies

Reference example, not the only acceptable wording: Teacher explanation: Ablation removes a component to investigate its contribution under a controlled comparison.

Interactions can make contributions depend on other components. Compare full system, without retrieval and without validation on identical cases and scoring.

Hold other settings fixed; record changed failure categories, not just totals.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of statistical tests and assumptions

Reference example, not the only acceptable wording: Teacher explanation: A statistical test compares observed **evidence** with a specified null model. A p-value is not the probability that the null hypothesis is true.

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed.

Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sampling variability

Reference example, not the only acceptable wording: Teacher explanation: Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 without replacement. Their means are1, 2, 2, 2, 2, 3.

Pictured end goal

Explain **sensitivity analysis** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Ablation studies

Statistical tests and assumptions

Sampling variability

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Sensitivity analysis** checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **sensitivity analysis**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach complete sensitivity patterns with explicit evaluation limits

> Distinguish retrieval-depth summaries, score decisions and measured correctness.

Definitions and complete contract

Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result.

Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.

Sensitivity item	Complete exercise contract
Main retrieval settings	k = 1, 3, 5; number of retrieved items, not a score threshold
Supplied recall summaries	0.4, 0.7, 0.8 at the respective settings
Supplied precision summaries	0.9, 0.7, 0.5 at the respective settings
Missing main information	Raw relevance labels, denominators and averaging convention; do not reconstruct counts
Separate score picture	Fixed scores 0.4, 0.6, 0.8; classify positive when score ≥ cutoff
Picture cutoffs	0.5, 0.7 and boundary check 0.6
Missing picture information	No reference labels; accuracy and best cutoff cannot be determined

Sensitivity item	Complete exercise contract
Interpretation limit	Only supplied settings are described; no beyond-range or universal guarantee

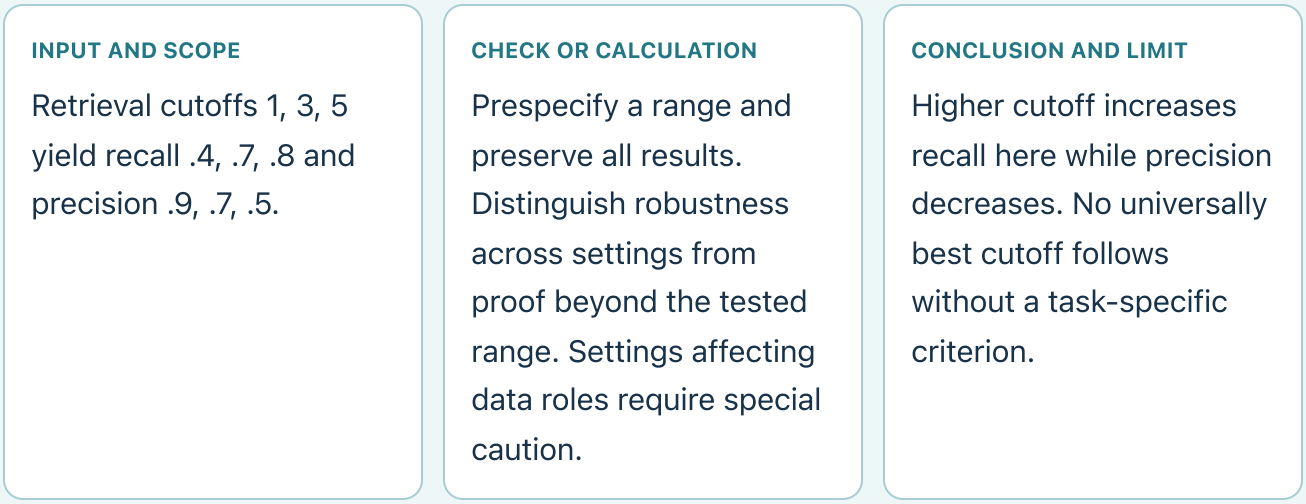
Main tradeoff and checked reasoning

Retrieved items k	Supplied recall	Supplied precision	Pattern in the tested range
1	0.4	0.9	Lowest recall; highest precision
3	0.7	0.7	Intermediate summaries
5	0.8	0.5	Highest recall; lowest precision
From k = 1 to k = 5	Increase 0.4, or 40 percentage points	Decrease 0.4, or 40 percentage points	Tradeoff; no universal best setting
What remains unknown	Raw labels/counts	Averaging convention	No invented per-query count or significance

Main retrieval depths 1, 3 and 5 have supplied recall 0.4, 0.7 and 0.8 while precision falls from 0.9 through 0.7 to 0.5. These are stipulated summaries with unspecified aggregation; do not invent raw labels. Picture flags are negative/positive/positive at cutoffs 0.5 and 0.6, and negative/negative/positive at 0.7. Equality is included. No reference labels are supplied, so no accuracy or best cutoff is known.

Independent thresholds 0.4, 0.5 and 0.6 give 8, 9 and 7 passes out of ten: range 7–9 passes, or 70%–90%, a twenty-percentage-point spread. The middle setting has the highest supplied count, but selecting it uses those outcomes; retain every result and evaluate a chosen setting separately. Do not infer a monotone pattern from that independent example.

Sensitivity analysis — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Qualification: A stable tested range does not establish stability under every parameter or dataset change.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.

Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Selection, equality and scope probes

Sensitivity audit	Checked interpretation
Report only the best tested setting	Hides sensitivity and outcome-driven selection
More positive predictions	Does not imply more correct predictions
Score equals cutoff	Positive under the stated inclusive rule
Raw scores unchanged	Decisions can still change when the rule changes
Recall increases as precision falls	Describe tradeoff; objective and constraints are needed for selection
Aggregate precision/recall supplied	Do not invent per-query counts or a missing averaging method
A pattern holds at three settings	Does not prove all intermediate or untested settings behave identically
Choose using these results	Disclose tuning and preserve separately reserved evaluation
Change data roles or split during sensitivity analysis	Record the change; leakage can invalidate comparison

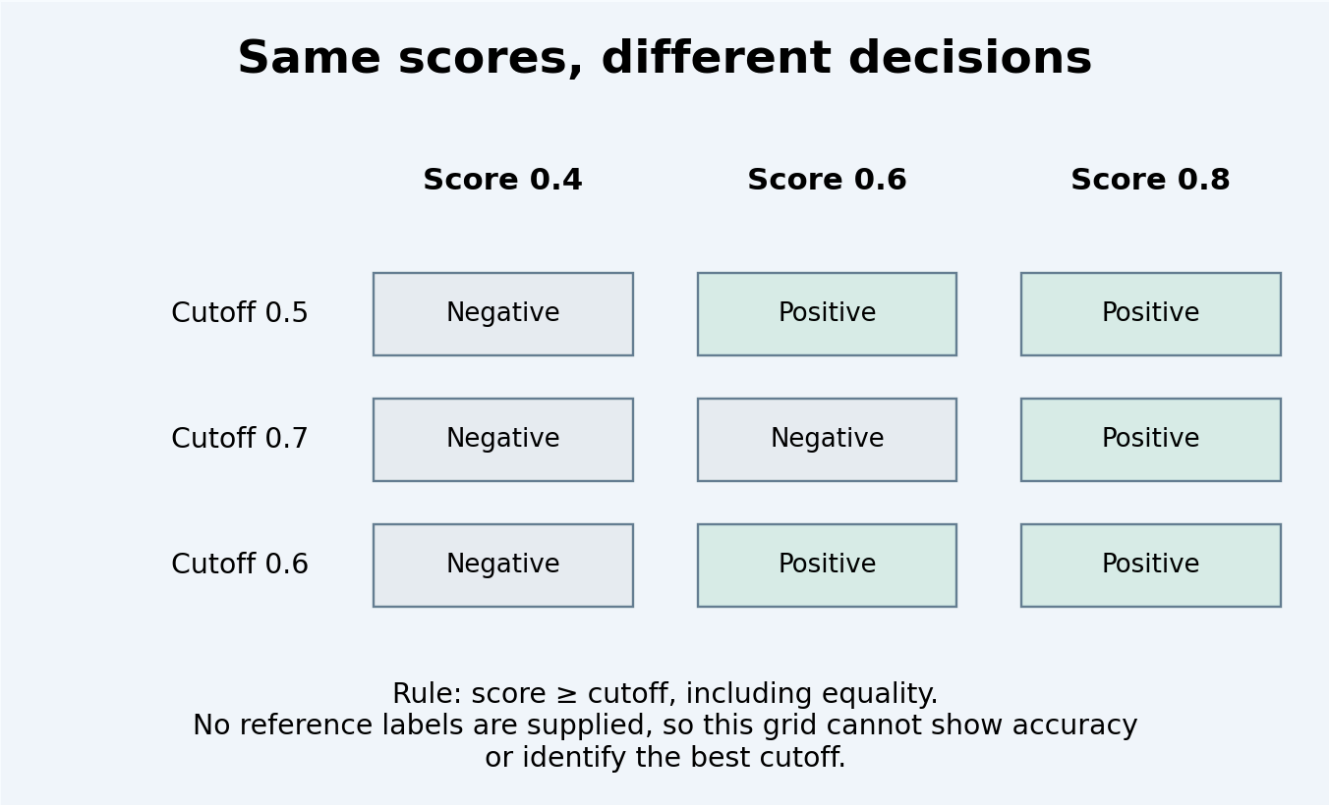
Separate fixed-score case and native grid

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

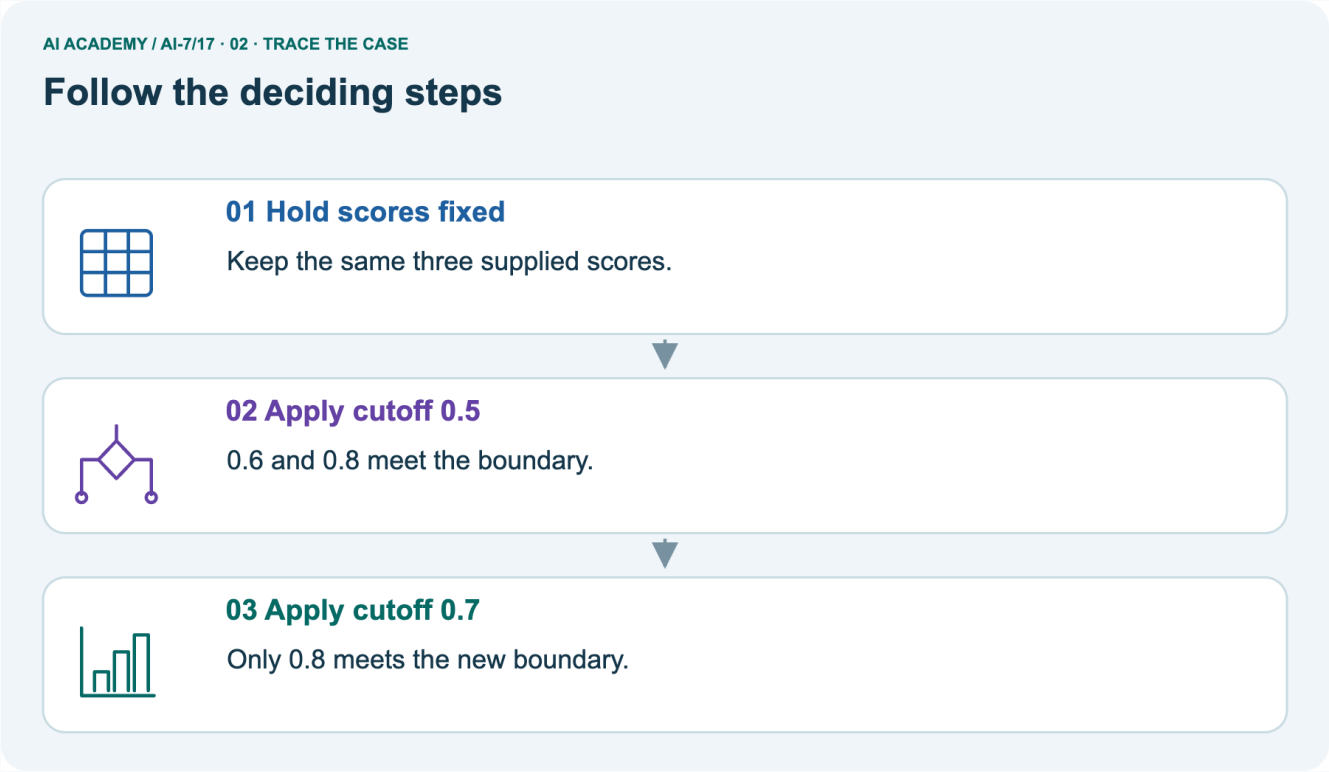
The picture fixes three unlabelled scores at 0.4, 0.6 and 0.8. The rule is positive when score is at least the cutoff. At 0.5 there are two positives; at 0.7 there is one; at exactly 0.6 there are two because equality counts. These are prediction flags, not correctness judgments.

If the unknown labels were negative/positive/positive, the first rule would fit all three; if instead negative/negative/positive, the second would. Those are explicitly hypothetical alternatives, not hidden reference labels. Without actual labels or a task objective, no best cutoff follows.



Fixed score	Cutoff 0.5	Cutoff 0.7	Cutoff 0.6
0.4	Negative	Negative	Negative
0.6	Positive	Negative	Positive: equality included
0.8	Positive	Positive	Positive
Positive count	2 of 3	1 of 3	2 of 3
Correct count	Unknown	Unknown	Unknown
Scores themselves	Unchanged	Unchanged	Unchanged

Picture trace and equality explanation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Hold scores fixed	Keep the same three supplied scores.
2. Apply cutoff 0.5	0.6 and 0.8 meet the boundary.
3. Apply cutoff 0.7	Only 0.8 meets the new boundary.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Without reference labels, these counts do not identify the best cutoff or establish correctness.

Change: Set the cutoff to exactly 0.6.

Prediction: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Sensitivity analysis	Checking outcomes across declared plausible choices
Robustness range	The conditions over which a conclusion is examined
Post-hoc selection	Choosing a procedure after seeing its scores

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sensitivity analysis

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/17

Sensitivity analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Hold scores fixed

Keep the same three supplied scores.



2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Sensitivity analysis**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A stable tested range does not establish stability under every parameter or dataset change.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Keep the same three supplied scores. 0.6 and 0.8 meet the boundary. Only 0.8 meets the new boundary. Changed case: 0.6 and 0.8 are positive because the rule includes equality.

The declared at-least boundary includes the threshold value. At cutoff 0.6, equality passes; absent reference labels, no best cutoff is established.

Changed-case reasoning: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main retrieval summaries show increasing recall and decreasing precision across the tested depths, with no raw-count reconstruction or universal best. Picture positive flags include equality and do not establish accuracy without labels. Independent pass counts 8, 9, 7 out of ten range from 70% to 90%, a twenty-point spread.

Threshold 0.5 has the highest supplied count but selecting it on these outcomes requires disclosed tuning and a fresh reserved evaluation.

Diagnose the current misconceptions

The worked result establishes every future case: A stable tested range does not establish stability under every parameter or dataset change.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

Teacher explanation: Rates**80%**, 90%, **70%**, range70–90%. Choosing .5 from these results is development selection, not independent final **evidence**.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Sensitivity analysis** checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result. Prespecify a range and preserve all results.

Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.

W2 Trace the worked case

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

W3 Audit the limit

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A stable tested range does not establish stability under every parameter or dataset change.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A stable tested range does not establish stability under every parameter or dataset change.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ **Earlier wording, alternatives and picture transcripts**

Guided and exit questions

Guided: Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Qualification: A stable tested range does not establish stability under every parameter or dataset change.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

Teacher explanation: Rates**80%**, 90%, **70%**, range70–90%. Choosing .5 from these results is development selection, not independent final **evidence**.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Rates**80%**, 90%, **70%**, range70–90%. Choosing .5 from these results is development selection, not independent final **evidence**. Require intermediate reasoning and the stated scope.

Fresh retry: Three settings score6, 7, 6 out of8. Expected reasoning: Rates75%, 87.5%, 75%; preserve all and note sensitivity. Reserved final data remains needed after selection.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **sensitivity analysis evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sensitivity analysis**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Dataset and benchmark limitations. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/18, ai-7/20, ai-7/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **sensitivity analysis**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Check every learner before discussion: Use the worked case for **Sensitivity analysis**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A stable tested range does not establish stability under every parameter or dataset change.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Set the cutoff to exactly 0.6. Require a **prediction** and a reason before comparing. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

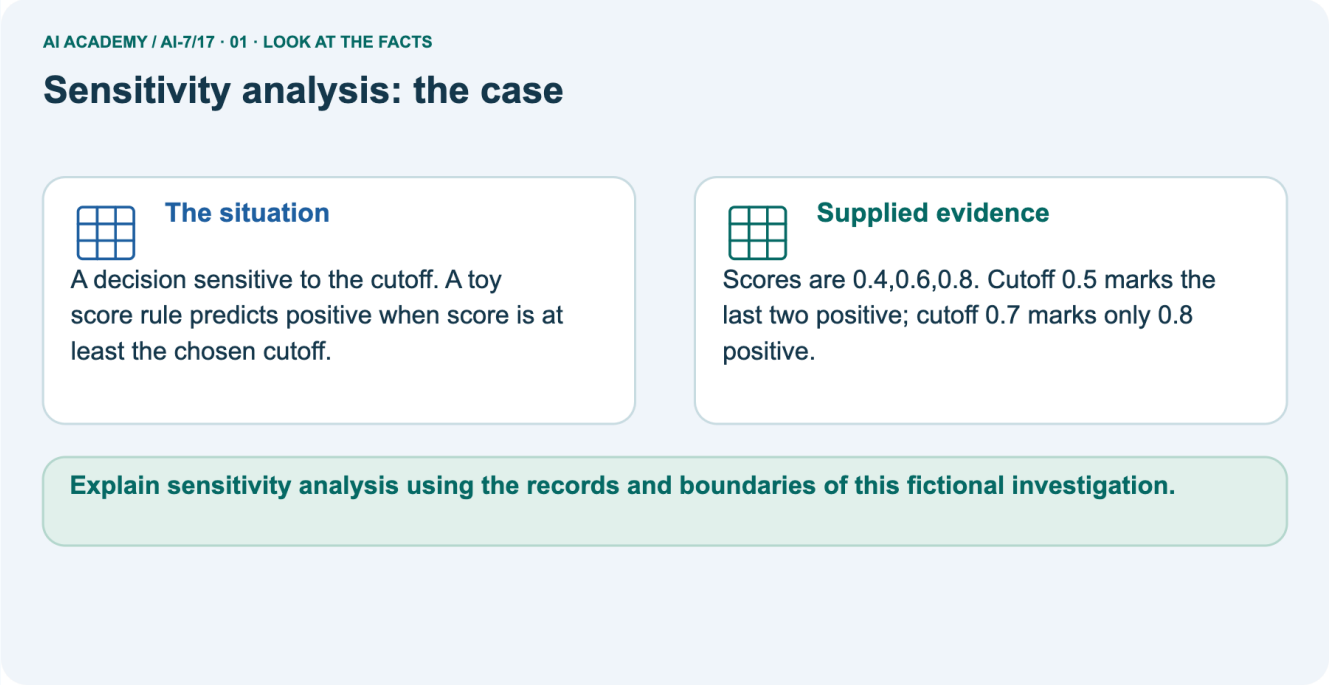
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Sensitivity analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

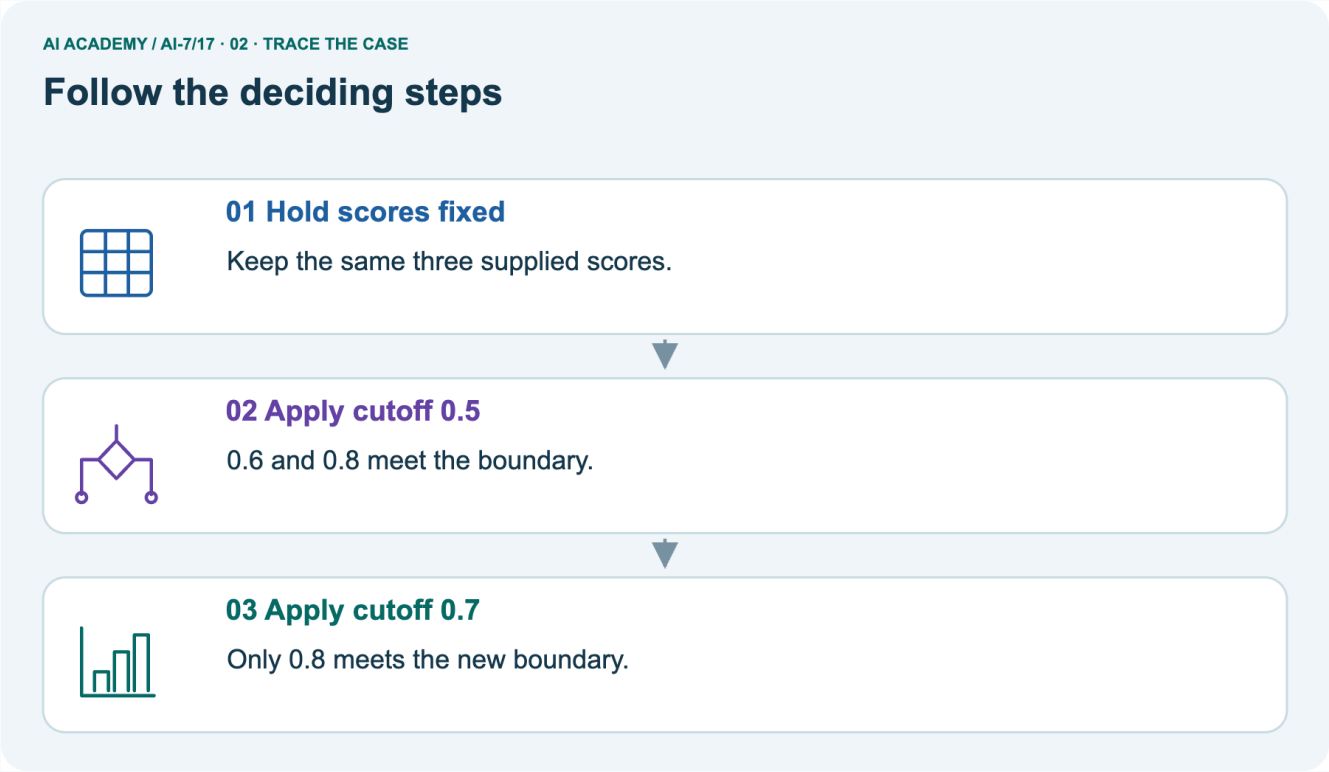
A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Supplied facts: Scores are 0.4,0.6,0.8.

Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



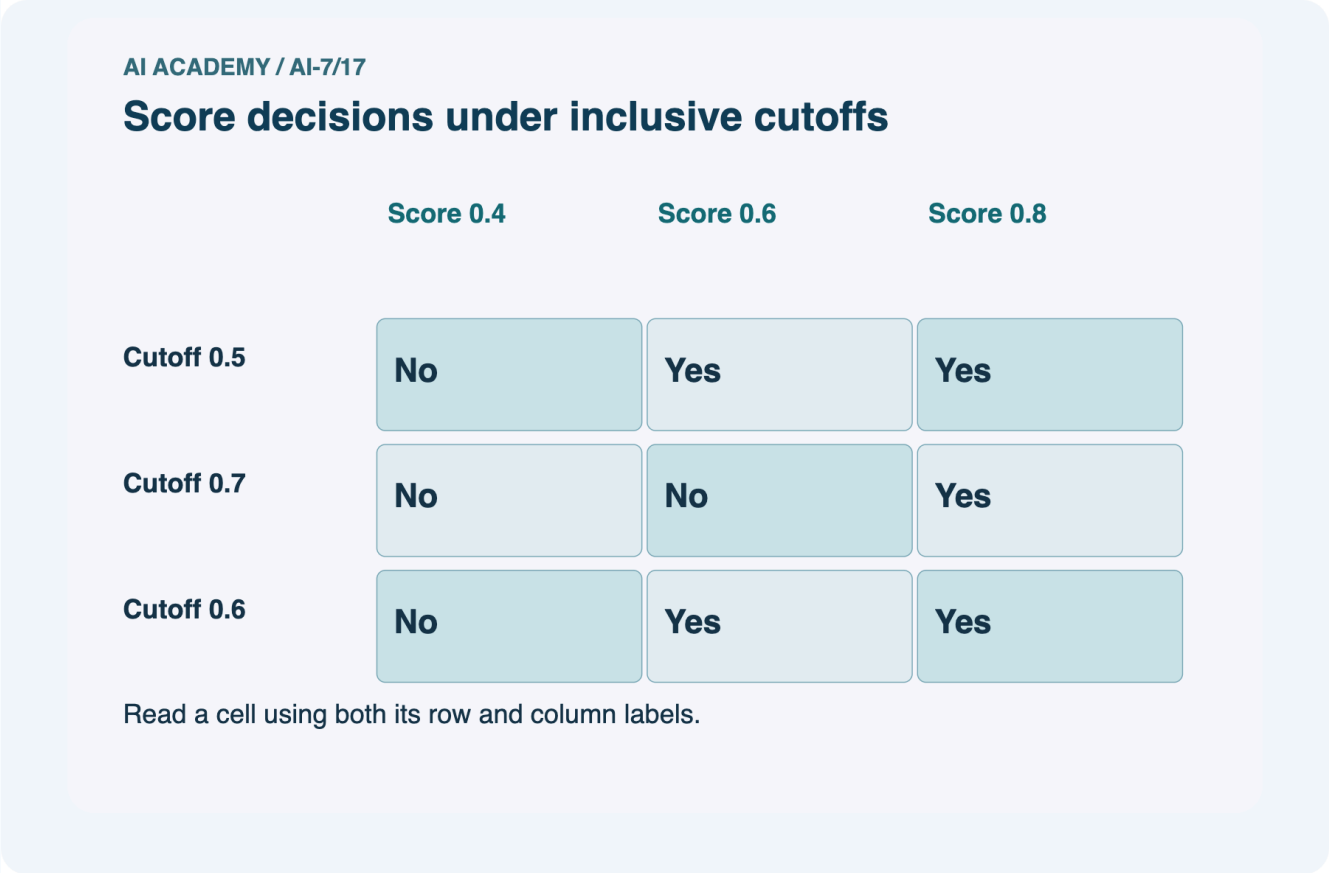
Follow the labelled picture with the matching explanation. Open the vector diagram

Hold scores fixed: Keep the same three supplied scores. → Apply cutoff 0.5: 0.6 and 0.8 meet the boundary. → Apply cutoff 0.7: Only 0.8 meets the new boundary.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Score decisions under inclusive cutoffs



Follow the labelled picture with the matching explanation. Open the vector diagram

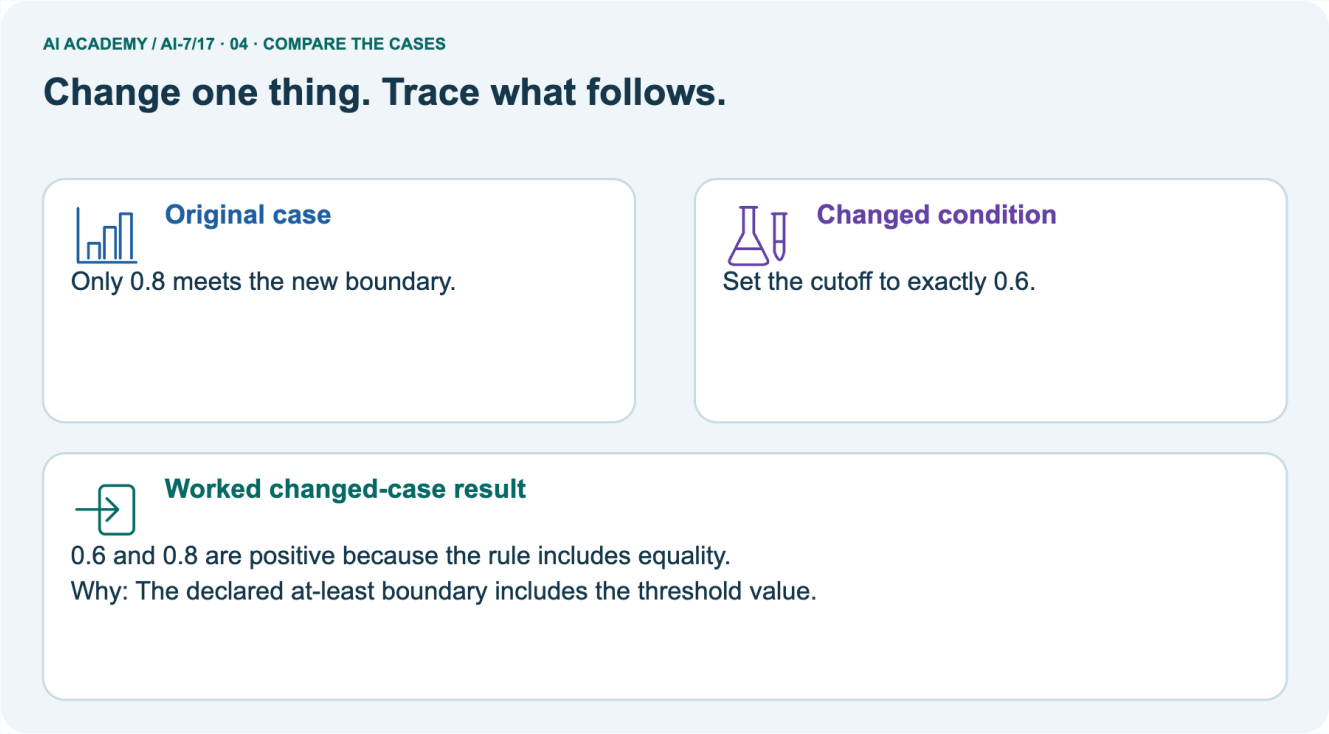
Score decisions under inclusive cutoffs. Cutoff 0.5: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes. Cutoff 0.7: Score 0.4 = No, Score 0.6 = No, Score 0.8 = Yes. Cutoff 0.6: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.8 meets the new boundary. Changed condition: Set the cutoff to exactly 0.6. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

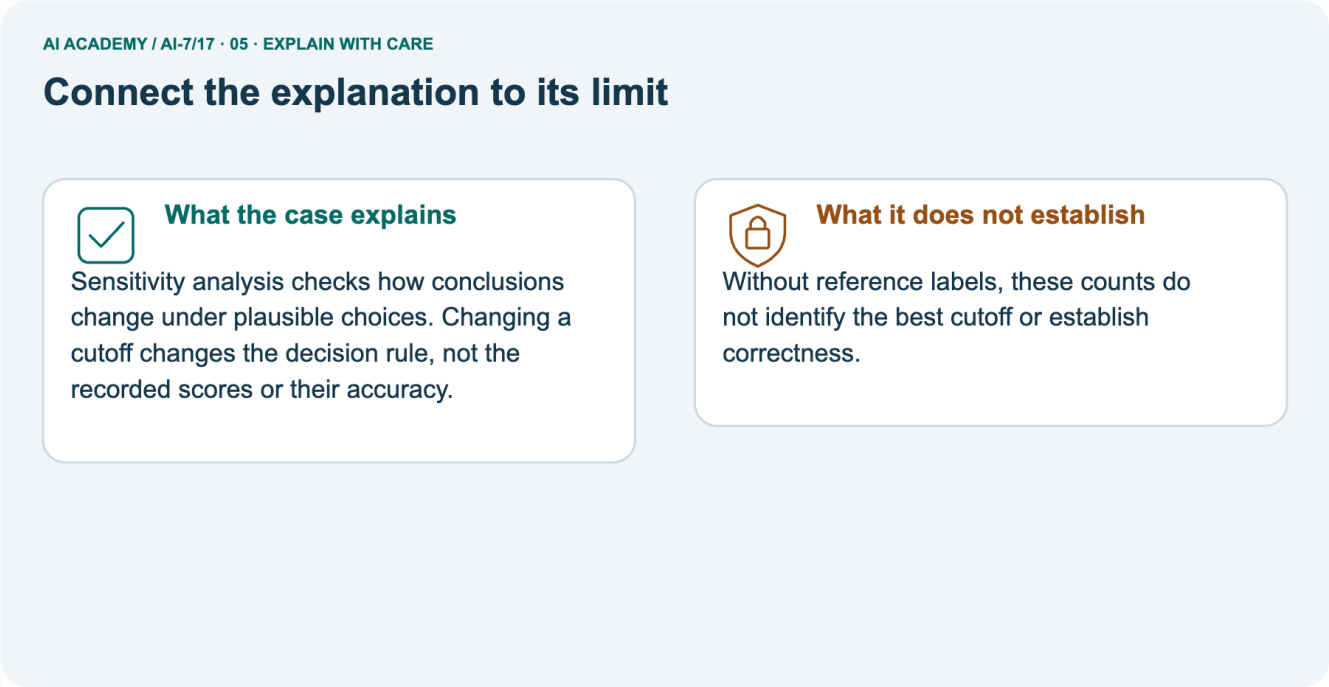
Reason: The declared at-least boundary includes the threshold value.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The declared at-least boundary includes the threshold value.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

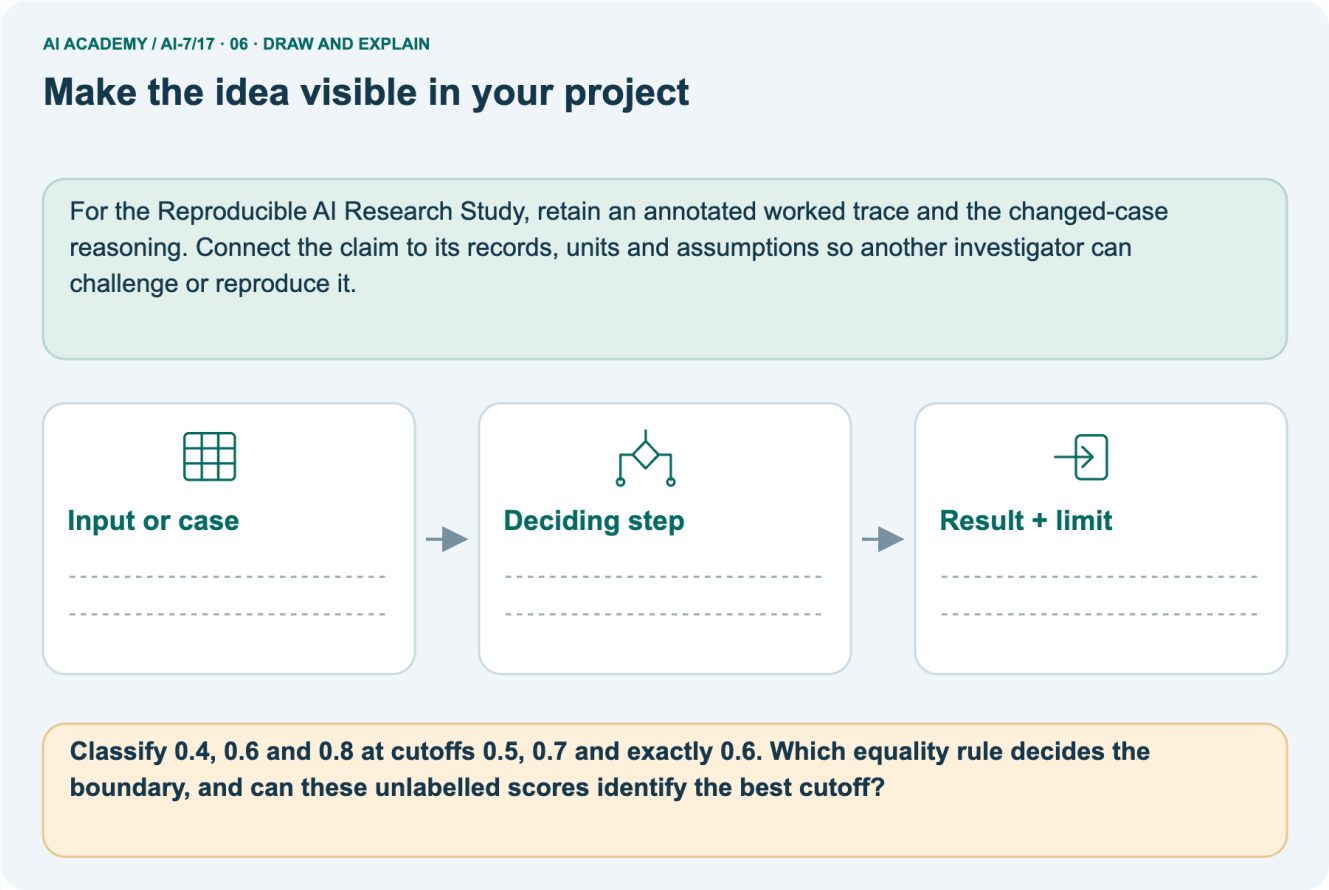
Limit: Without reference labels, these counts do not identify the best cutoff or establish correctness.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Without reference labels, these counts do not identify the best cutoff or establish correctness.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep the same three supplied scores. 0.6 and 0.8 meet the boundary. Only 0.8 meets the new boundary. Changed case: 0.6 and 0.8 are positive because the rule includes equality.

The declared at-least boundary includes the threshold value. At cutoff 0.6, equality passes; absent reference labels, no best cutoff is established.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sensitivity analysis: the case

02. Follow the deciding steps

03. Score decisions under inclusive cutoffs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.